

Coffee Kiosk POS - Customer Frontend

Functional Requirements Specification

Document field	Value
Project	Coffee Kiosk POS
Module	Customer - Frontend
Document type	Functional Requirements Specification (FRS)
Author	Kushagra Sinha
Version	1.0
Date	10 September 2026
Status	Draft for review

Table of Contents

- [1. Version History](#)
- [2. Introduction](#)
 - [2.1 Purpose of Document](#)
 - [2.2 Project Scope](#)
 - [2.3 Definitions and Abbreviations](#)
 - [2.4 System Overview](#)
 - [2.5 Customer Order Flow](#)
- [3. User Personas, Roles, and Permissions](#)
- [4. Functional Requirements](#)
 - [4.1 Requirement Priority](#)
 - [4.2 Food Catalog](#)
 - [4.3 Cart](#)
 - [4.4 Checkout](#)
 - [4.5 Payment Gateway](#)
 - [4.6 Order Confirmation and Receipt](#)
- [5. Business Rules and Assumptions](#)
- [6. Non-Functional Requirements](#)
 - [6.1 Security and Session Management](#)
 - [6.2 Privacy](#)
 - [6.3 Usability and Accessibility](#)
 - [6.4 Reliability and Compatibility](#)
- [7. Sign-off](#)

The interface images in this document are illustrative mockups. The written requirements are the source of truth for future test scenarios, test cases, RTM, execution, and defect reporting.

1. Version History

Version	Date	Description of Changes	Author	Status
1.0	25 September 2026	Initial FRS for the Coffee Kiosk POS customer-facing frontend	Kushagra Sinha	Draft for review

2. Introduction

2.1 Purpose of Document

This document defines the customer-facing frontend functionality of Coffee Kiosk POS. It describes the expected behaviour from menu browsing through payment and receipt generation so that later STLC documents can be prepared against stable, testable requirements.

2.2 Project Scope

Coffee Kiosk POS is a self-ordering application for retail coffee stores. A customer uses an in-store touchscreen kiosk to:

- Browse the available food and beverage menu.
- Add available items to a cart and review the order.
- Choose dine-in or takeaway.
- Apply a valid store coupon, when available.
- Select UPI or Card as the payment method.
- Complete payment through a third-party payment gateway.
- View a bill receipt and optionally receive it by email.

This FRS covers only the customer-facing frontend. Brew Crew operations, administrative settings, backend service behaviour, database testing, and direct testing of the third-party payment provider are outside its implementation scope. Their customer-visible results and integration points remain in scope.

2.3 Definitions and Abbreviations

Term	Definition
AWS	Amazon Web Services; the managed cloud platform used for application data services.
Brew Crew	Store employees who manage item availability, coupons, and order fulfilment.
Card	Debit or credit card payment handled by the third-party payment gateway.
FRS	Functional Requirements Specification.
IT Admin	Technical staff responsible for application services and supporting infrastructure.
POS	Point of Sale.
Session	The temporary customer order state held while one kiosk order is in progress.
STLC	Software Testing Life Cycle.
UI	User Interface.
UPI	Unified Payments Interface.

2.4 System Overview

Coffee Kiosk POS uses a ReactJS customer frontend connected to backend services developed with Spring Boot. The backend provides menu, price, coupon, order, and payment-status data. UPI and Card payments are handled by a third-party payment gateway. Application data is stored using an AWS-managed database service.

The kiosk remains lightweight because the customer device runs the frontend while core processing, data storage, and payment handling are managed by centralized services.

2.5 Customer Order Flow

Food Catalog → Cart → Checkout → Payment Gateway → Order Confirmation and Receipt

If payment fails or is cancelled, the customer returns to a recoverable payment state without receiving a successful receipt.

3. User Personas, Roles, and Permissions

Role	Purpose	Main permissions	Relevance to this FRS
Customer	Places an order using the self-service kiosk.	Browse available items, add or remove items, change quantities, choose dine-in or takeaway, apply a coupon, select UPI or Card, pay, view the receipt, and request an emailed receipt.	Primary and in scope.
Brew Crew	Operates the store and fulfils orders.	Manage item availability and valid coupons; search, filter, and view orders; change order status to Cancelled, Processing, Ready for Takeaway, Ready to Serve, or Completed; unlock a kiosk with a passcode; and manage the kiosk password through protected settings.	Interfaces are out of scope. Customer-visible availability, coupon results, and order status are in scope.
IT Admin	Maintains the technical platform.	Manage application services, payment-gateway integration, cloud services, monitoring, and related technical configuration.	Administrative interfaces are out of scope. Service availability and customer-visible integration failures are in scope.

The customer frontend shall not expose Brew Crew or IT Admin functions without the required protected access.

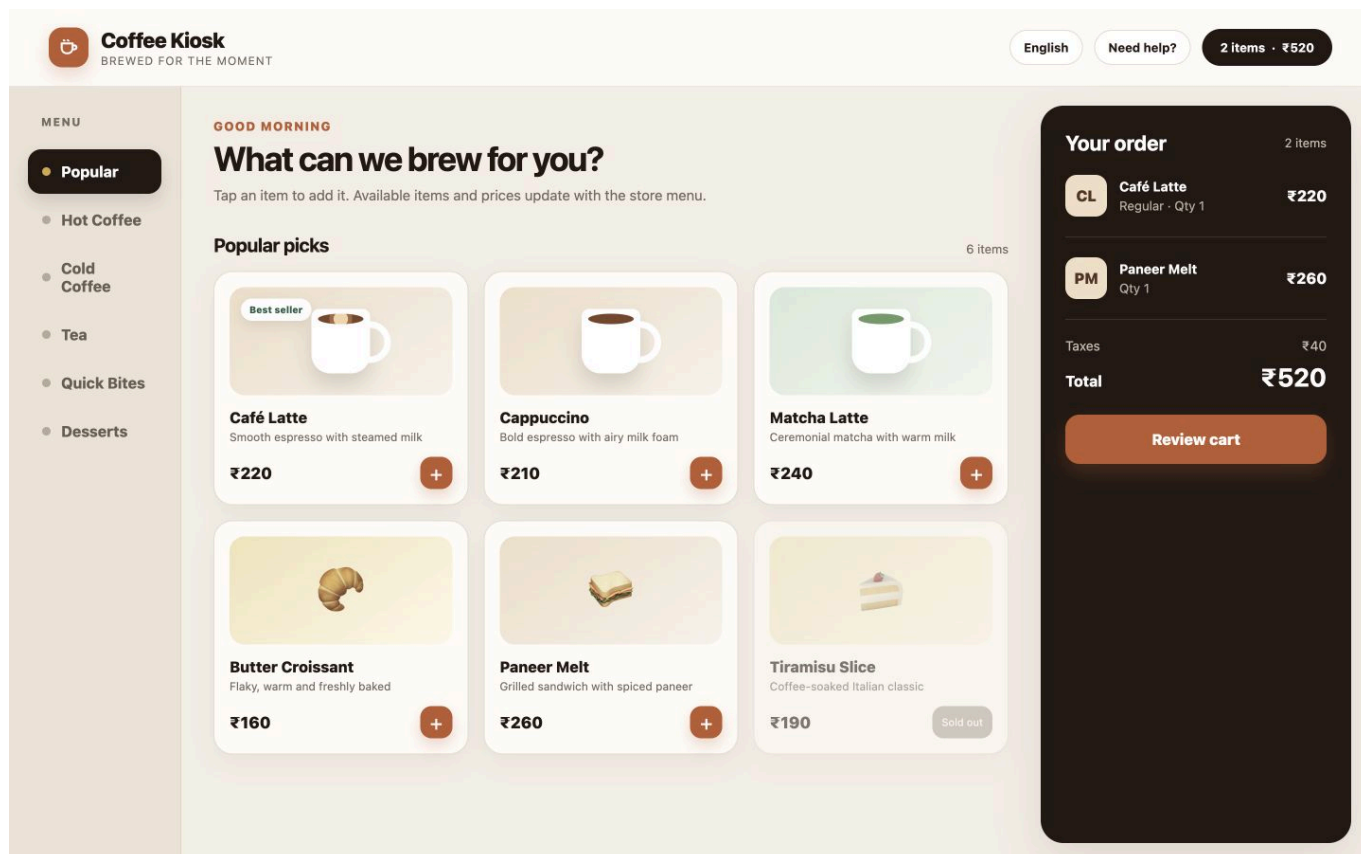
4. Functional Requirements

4.1 Requirement Priority

Priority	Meaning
P0	Essential to complete the customer order or payment flow.
P1	Important for a complete, safe, and usable release.
P2	Useful supporting behaviour that does not block the main flow.

4.2 Feature - Food Catalog

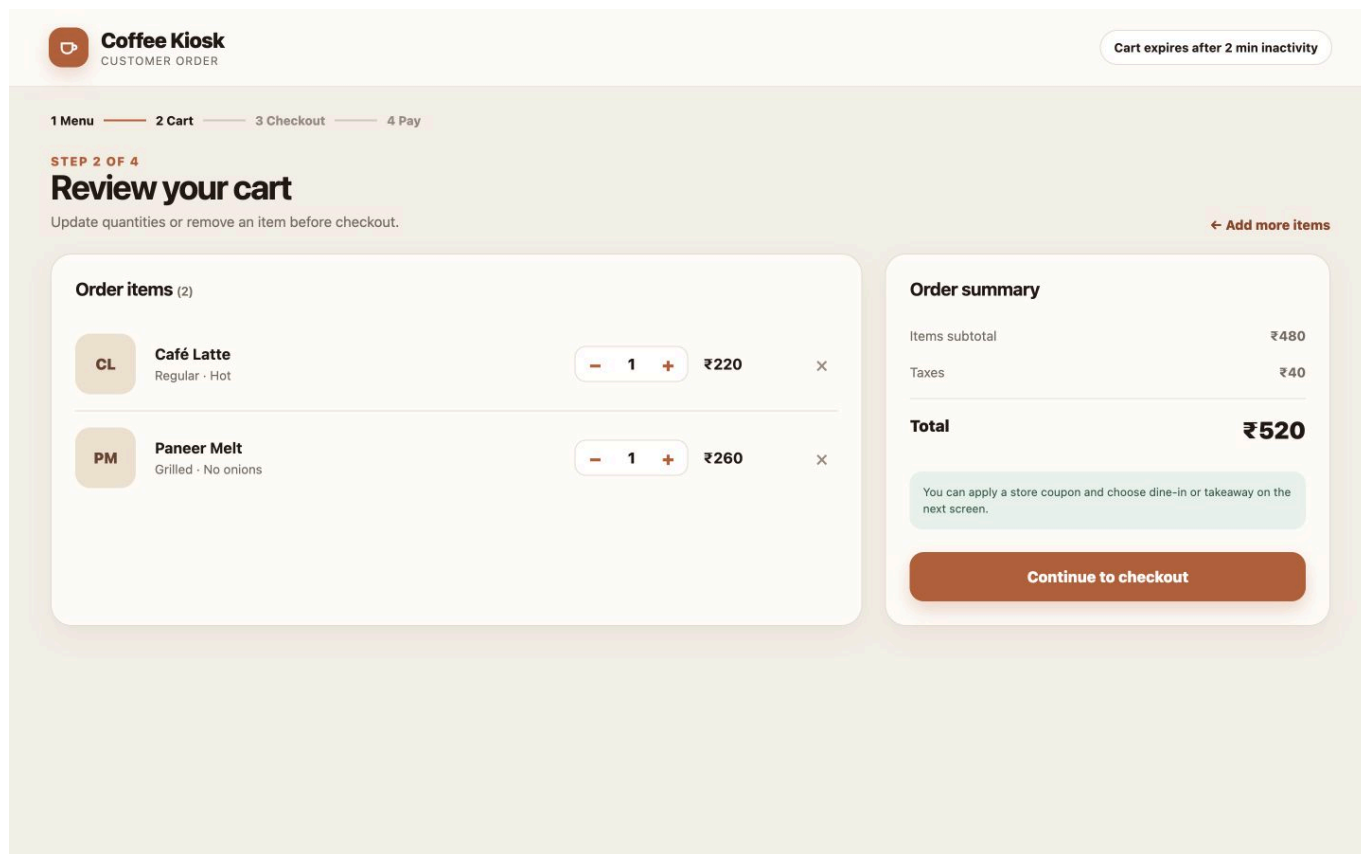
Requirement ID	Priority	Requirement
REQ-CAT-01	P0	The system shall open the customer session on a menu screen that shows the available item categories.
REQ-CAT-02	P0	The system shall show each menu item with its name, image, short description, and current price.
REQ-CAT-03	P0	The system shall clearly identify unavailable items and shall not allow them to be added to the cart.
REQ-CAT-04	P1	The customer shall be able to switch between menu categories without losing items already added to the cart.
REQ-CAT-05	P1	When an item has required store-defined choices, such as size or serving option, the system shall require those choices before adding the item.
REQ-CAT-06	P0	The customer shall be able to add an available item to the cart and receive visible confirmation that the cart count and total have changed.
REQ-CAT-07	P1	The menu screen shall provide a visible path to review the cart at any time.



Illustrative Food Catalog showing categories, product availability, prices, and a live cart summary.

4.3 Feature - Cart

Requirement ID	Priority	Requirement
REQ-CART-01	P0	The cart shall list every selected item with its chosen options, quantity, and line amount.
REQ-CART-02	P0	The customer shall be able to increase or decrease an item's quantity using touch controls.
REQ-CART-03	P0	The customer shall be able to remove an item from the cart.
REQ-CART-04	P0	The system shall recalculate the subtotal, applicable taxes, and total whenever the cart changes.
REQ-CART-05	P1	The cart shall provide options to continue browsing the menu or proceed to checkout.
REQ-CART-06	P1	When the cart is empty, the system shall show an empty-cart message and shall not allow checkout to begin.



Illustrative Cart showing item details, quantity controls, removal controls, and calculated totals.

4.4 Feature - Checkout

Requirement ID	Priority	Requirement
REQ-CHK-01	P0	The checkout screen shall show a locked final order list; item and quantity changes shall be made by returning to the cart.
REQ-CHK-02	P0	The checkout screen shall show the item subtotal, taxes, coupon discount when applicable, and final amount to be paid.
REQ-CHK-03	P0	The customer shall be required to select exactly one order type: Dine-in or Takeaway.
REQ-CHK-04	P1	The customer shall be able to enter an optional store coupon code and request validation.
REQ-CHK-05	P0	For a valid coupon, the system shall show the applied discount and recalculate the payable amount. For an invalid, expired, or inapplicable coupon, it shall show a clear message and apply no discount.
REQ-CHK-06	P0	The customer shall be required to select exactly one payment method: UPI or Card.
REQ-CHK-07	P0	The payment action shall use the locked order and displayed payable amount and shall remain unavailable until the required checkout choices are complete.

Requirement ID	Priority	Requirement
REQ-CHK-08	P1	Returning to the cart shall allow order changes, and the checkout totals shall be recalculated when the customer enters checkout again.

The screenshot shows the 'Confirm and pay' checkout screen for 'Coffee Kiosk'. The top navigation bar includes '1 Menu', '2 Cart', '3 Checkout', and '4 Pay'. The current step is 'STEP 3 OF 4 Confirm and pay'. A message states: 'Your order is locked for review. Return to cart to make changes.' with a 'Back to cart' link.

Final order (Locked for checkout):

- 1 x Café Latte (Regular - Hot) ₹220
- 1 x Paneer Melt (Grilled - No onions) ₹260
- Items subtotal ₹480
- Taxes ₹40
- Coupon - BREW10 - ₹52
- Amount to pay ₹468**

Order type: ☒ Dine-in, ☐ Takeaway

Store coupon: BREW10 (Apply button). Status: ✓ Coupon applied - You saved ₹52

Payment method: ☒ UPI, ☐ Card

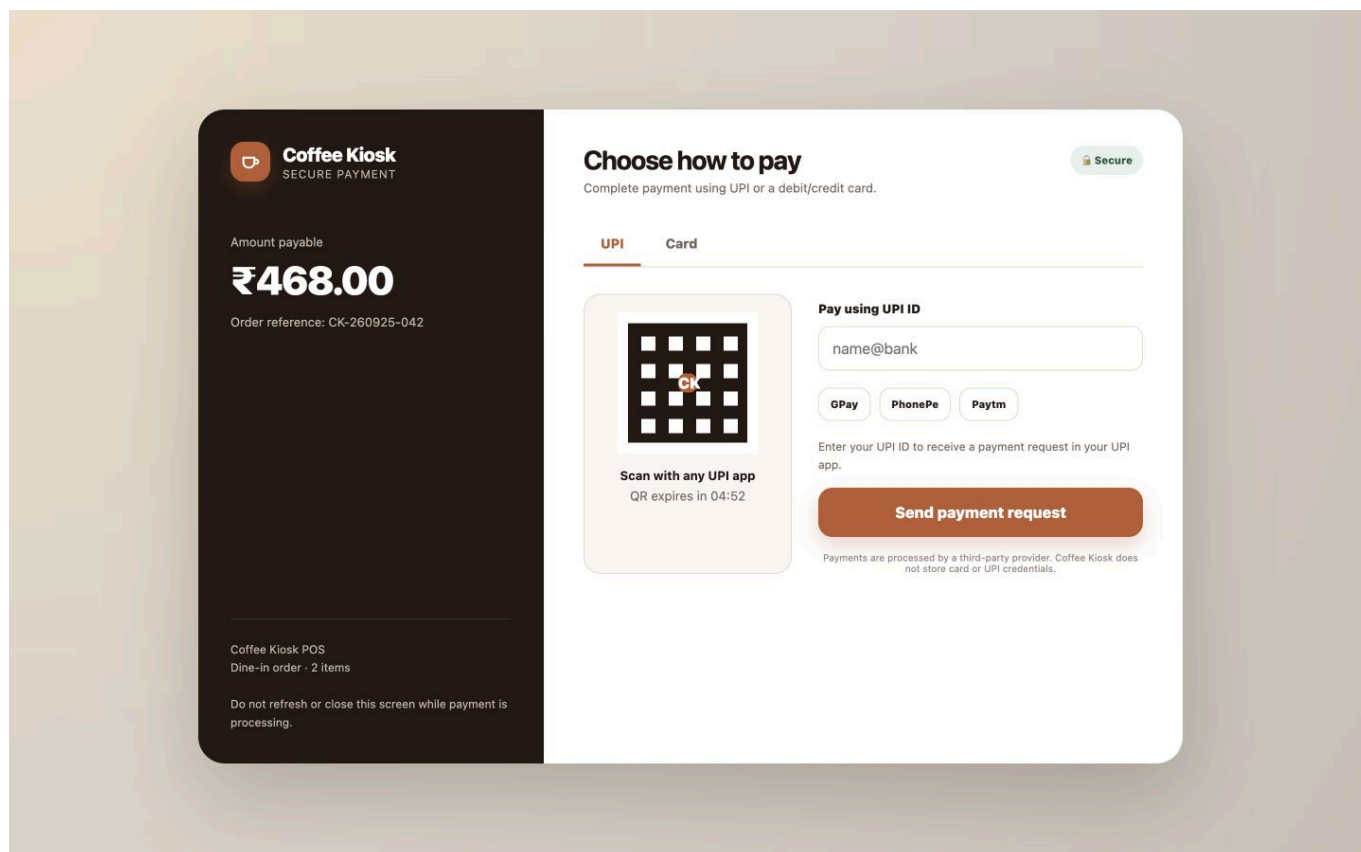
Pay ₹468 securely (button). Below the button: 'You will continue to our secure payment partner.'

Illustrative Checkout showing the locked order, dine-in/takeaway selection, coupon result, and payment-method selection.

4.5 Feature - Payment Gateway (UPI and Card)

Requirement ID	Priority	Requirement
REQ-PAY-01	P0	The payment-method screen shall allow the customer to navigate between UPI and Card payment options.
REQ-PAY-02	P0	The payment screen shall display the merchant name, order reference, and exact amount to be paid.
REQ-PAY-03	P0	When UPI is selected, the gateway shall present the available UPI payment options and validate the information required by the selected option.
REQ-PAY-04	P0	When Card is selected, the gateway shall present the required Card fields and validation through the third-party payment service.
REQ-PAY-05	P0	After payment is submitted, the system shall show a processing state and prevent repeated payment submission.

Requirement ID	Priority	Requirement
REQ-PAY-06	P0	The system shall show a successful order confirmation only after it receives a verified successful payment result.
REQ-PAY-07	P0	If payment fails, times out, or is cancelled, the system shall show a clear result and allow the customer to retry or choose another supported payment method without rebuilding the cart.

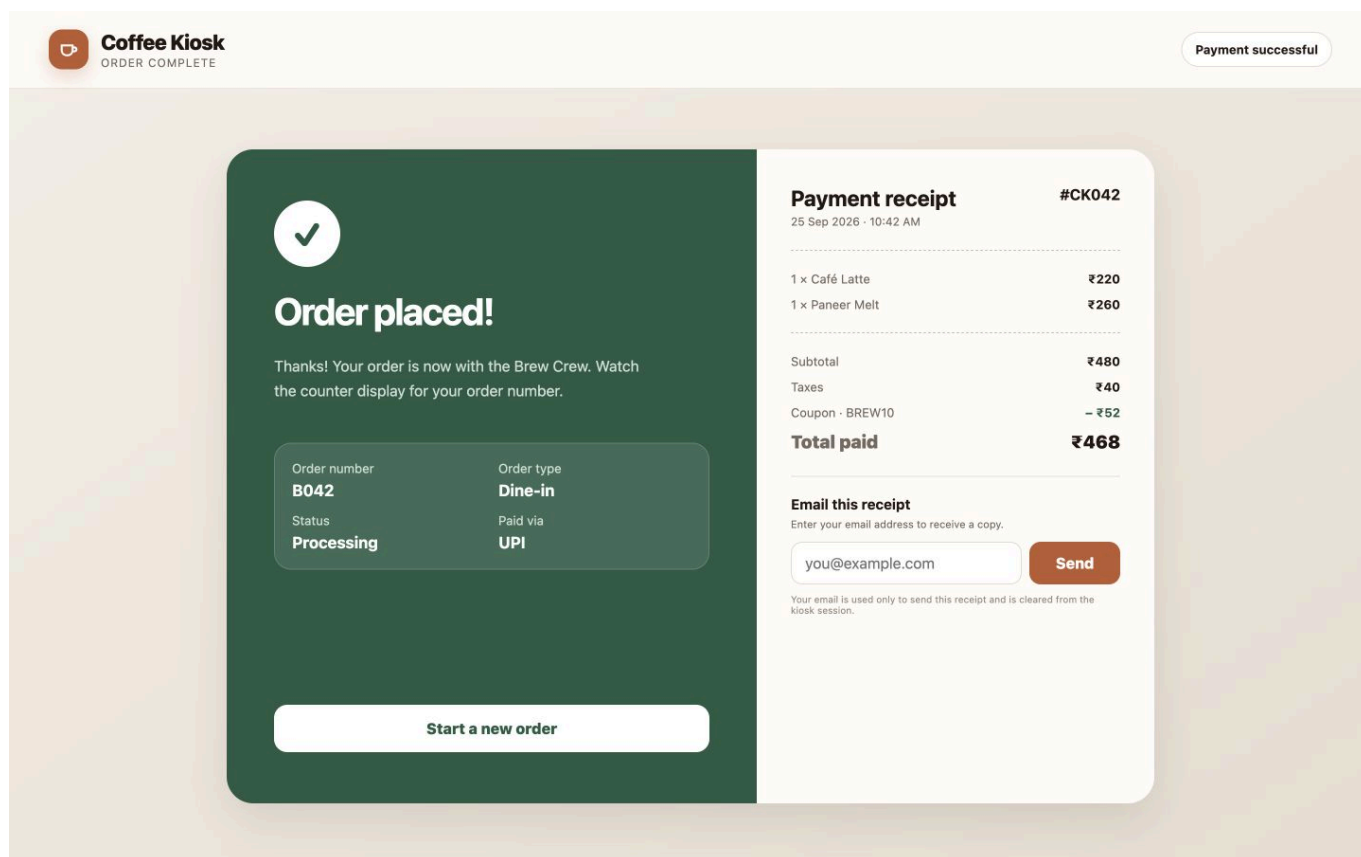


Illustrative third-party payment screen showing the payable amount, order reference, and UPI/Card navigation.

4.6 Feature - Order Confirmation and Receipt

Requirement ID	Priority	Requirement
REQ-RCT-01	P0	After successful payment, the system shall show an order-confirmation screen with the order number, order type, initial order status, and payment method.
REQ-RCT-02	P0	The system shall show an itemised bill receipt containing the ordered items, subtotal, taxes, coupon discount when applicable, and total paid.
REQ-RCT-03	P1	The customer shall be able to enter an email address and request a copy of the receipt.

Requirement ID	Priority	Requirement
REQ-RCT-04	P1	The system shall validate the email format and show whether the receipt request was sent or could not be completed.
REQ-RCT-05	P0	The customer shall be able to start a new order, which shall clear the previous cart and customer session from the kiosk.



Illustrative success screen with the order number, itemised bill, and optional emailed receipt.

5. Business Rules and Assumptions

ID	Rule or assumption
BR-01	Customer orders are placed as guest sessions; customer account registration and login are not included.
BR-02	Menu items, prices, taxes, availability, and coupon rules are supplied by backend services and store configuration.
BR-03	UPI and Card are the supported customer payment methods for this release; cash payment is not included.
BR-04	One valid coupon may be applied to an order at a time.
BR-05	Prices and payment amounts are displayed in Indian Rupees (INR).

ID	Rule or assumption
BR-06	Sending the receipt by email is optional and is not required to complete an order.

6. Non-Functional Requirements

6.1 Security and Session Management

Requirement ID	Priority	Requirement
REQ-SEC-01	P1	If no touch input is detected for 2 continuous minutes on customer-controlled screens, the system shall clear the local session, including cart items, coupon data, and entered email, and return to the menu. An active payment-result check shall not be interrupted as an idle session.
REQ-SEC-02	P0	Brew Crew and IT Admin functions shall not be accessible from the customer flow without protected authorization.
REQ-SEC-03	P0	The customer frontend shall not store complete Card details, UPI credentials, or payment authentication data.
REQ-SEC-04	P1	Payment and coupon requests shall be sent only through the approved backend and third-party payment integrations.

6.2 Privacy

Requirement ID	Priority	Requirement
REQ-PRV-01	P1	An email address entered for a receipt shall be used only for that receipt request and shall be cleared from the kiosk session afterward.
REQ-PRV-02	P1	The kiosk shall not display one customer's order or email information after the session is cleared or a new order begins.

6.3 Usability and Accessibility

Requirement ID	Priority	Requirement
REQ-USE-01	P1	Primary controls shall be clearly labelled, touch-friendly, and visually consistent across the customer flow.
REQ-USE-02	P1	The system shall provide visible feedback when an item is added, a coupon is checked, payment is processing, or an action cannot be completed.
REQ-USE-03	P1	Error messages shall explain the problem in simple language and identify the next available customer action.

Requirement ID	Priority	Requirement
REQ-USE-04	P2	Important states shall use text or icons in addition to colour so they remain understandable without relying on colour alone.

6.4 Reliability and Compatibility

Requirement ID	Priority	Requirement
REQ-REL-01	P0	The cart and checkout selections shall remain consistent while the customer moves through the normal order flow or retries a failed payment.
REQ-REL-02	P0	Repeated taps on checkout or payment actions shall not create duplicate orders or duplicate payment requests.
REQ-REL-03	P0	A connectivity or service error shall not be shown as a successful order or successful payment.
REQ-CMP-01	P1	The customer frontend shall render correctly at the approved landscape kiosk resolution in the store's supported modern Chromium-based browser.
REQ-PERF-01	P1	While waiting for menu, coupon, order, or payment responses, the frontend shall show a visible loading or processing state and prevent accidental repeated submission.

7. Sign-off

Approval confirms that this FRS is a suitable baseline for the Coffee Kiosk POS customer-facing frontend. Material scope changes after approval should be reviewed for their effect on future STLC documents.

Stakeholder	Representative Name and Designation	Approval Method (JIRA/Confluence/Email)	Date Approved	Current Status
QA Department	Kushagra Sinha (QA Engineer)	Email	-	Pending approval